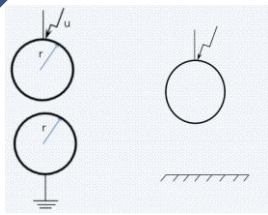
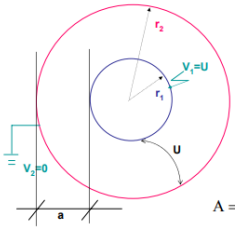
EE4047 – High Voltage Techniques I

Continuation to:
Cylindrical Electrode Systems





Week 10.1 – Monday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

08.05.2023

OUTLINE

2 Multilayer Dielectric Concentric Spheres

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Multilayer dielectric materials between the electrodes:

$$I = I_1 = I_2 = I_3 = \cdots = I_n$$

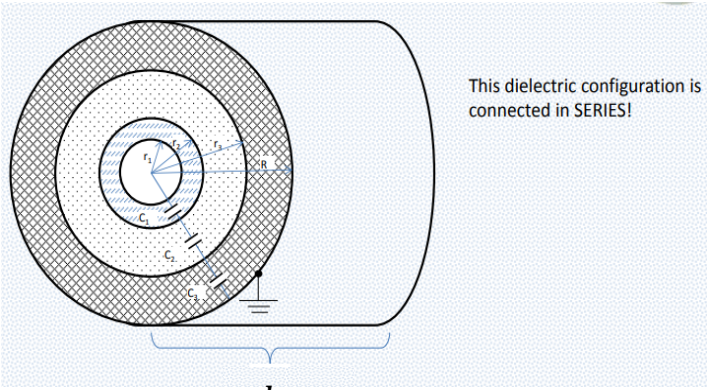
$$Q = Q_1 = Q_2 = Q_3 = \cdots = Q_n$$

$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U = U_1 + U_2 + U_3 + \cdots + U_n$$

U is the voltage applied to the entire system

U_1, U_2, …, U_n are voltages across the corresponding dielectric layers



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Voltages across each dielectric layer:

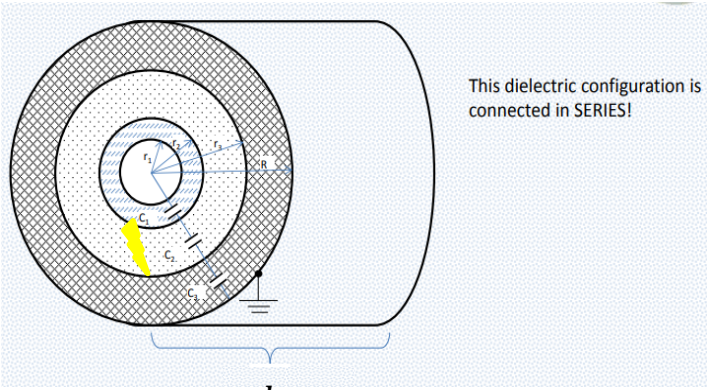
$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U_1 = \frac{C\,U}{C_1}$$

$$U_2 = \frac{C\,U}{C_2}$$

$$\vdots$$

$$U_n = \frac{C\,U}{C_n}$$



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

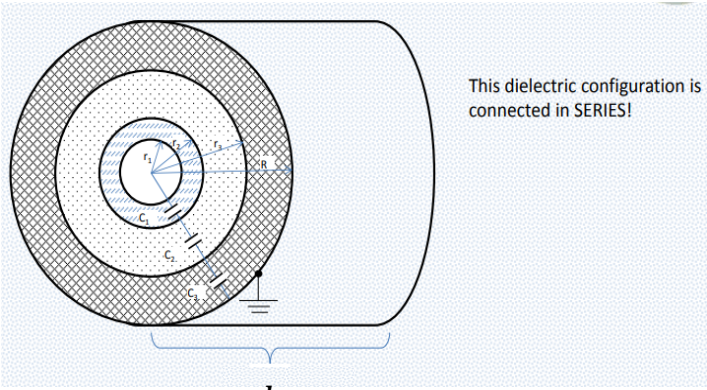
Equivalent capacitance of the system:

$$X_C = X_{C1} + X_{C2} + X_{C3} + \cdots + X_{Cn}$$

$$\frac{1}{\omega C} = \frac{1}{\omega C_1} + \frac{1}{\omega C_2} + \frac{1}{\omega C_3} + \cdots + \frac{1}{\omega C_n}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$$

C : equivalent capacitance of the system, total capacitance of the system
U : voltage applied to the entire system
C_i : capacitance of the ith dielectric layer
U_i : voltage across i^{th} dielectric layer



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Calculating capacitance:

$$C_1 = 2\pi\varepsilon_1 l \frac{1}{\ln\frac{r_2}{r_1}}, C_2 = 2\pi\varepsilon_2 l \frac{1}{\ln\frac{r_3}{r_2}}, \dots, C_n = 2\pi\varepsilon_n l \frac{1}{\ln\frac{R}{r_n}}$$

$$\frac{1}{C} = \frac{1}{2\pi\varepsilon_1 l \frac{1}{\ln\frac{r_2}{r_1}}} + \frac{1}{2\pi\varepsilon_2 l \frac{1}{\ln\frac{r_3}{r_2}}} + \cdots + \frac{1}{2\pi\varepsilon_n l \frac{1}{\ln\frac{R}{r_n}}}$$

$$= \frac{1}{2\pi l}\left(\frac{1}{\varepsilon_1}\ln\frac{r_2}{r_1} + \frac{1}{\varepsilon_2}\ln\frac{r_3}{r_2} + \cdots + \frac{1}{\varepsilon_n}\ln\frac{R}{r_n}\right) = \frac{1}{2\pi l}\underbrace{\sum_{i=1}^{n}\frac{1}{\varepsilon_i}\ln\frac{r_{i+1}}{r_i}}_{A}$$

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Total capacitance, equivalent capacitance:

$$A = \sum_{i=1}^{n} \frac{1}{\varepsilon_i} \ln \frac{r_{i+1}}{r_i} = \frac{1}{\varepsilon_1} \ln \frac{r_2}{r_1} + \frac{1}{\varepsilon_2} \ln \frac{r_3}{r_2} + \cdots + \frac{1}{\varepsilon_n} \ln \frac{R}{r_n}$$

$$\frac{1}{C} = \frac{A}{2\pi l} \Rightarrow C = \frac{2\pi l}{A}$$

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Potantial and electric field across 1st dielectric layer:

$$U_1 = \frac{C\,U}{C_1} = \frac{\frac{2\pi l}{A}U}{2\pi\varepsilon_{\,1} l\,\frac{1}{\ln\frac{r_2}{r_1}}} = \frac{U}{A\varepsilon_{\,1}}\ln\frac{r_2}{r_1}$$

$$E_{1max} = \frac{U_1}{r_1 \ln\frac{r_2}{r_1}} = \frac{\frac{U}{A\varepsilon_{\,1}}\ln\frac{r_2}{r_1}}{r_1 \ln\frac{r_2}{r_1}} = \frac{U}{A\varepsilon_1 r_1}$$

$$E_{1min} = \frac{U_1}{r_2 \ln\frac{r_2}{r_1}} = \frac{\frac{U}{A\varepsilon_{\,1}}\ln\frac{r_2}{r_1}}{r_2 \ln\frac{r_2}{r_1}} = \frac{U}{A\varepsilon_1 r_2}$$

We can write the same equations with respect to applied voltage (U)

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Potantial and electric field across i th dielectric layer:

$$U_i = \frac{C.U}{C_i} = \frac{U}{A.\varepsilon_i.l_i}.\ln\frac{r_{i+1}}{r_i}$$

Voltage across i^{th} layer

Applied Voltage

$$E_{imax} = \frac{U_i}{r_i.\ln\frac{r_{i+1}}{r_i}} = \frac{U}{A.\varepsilon_i.r_i}$$

$$E_{imin} = \frac{U_i}{r_{i+1}.\ln\frac{r_{i+1}}{r_i}} = \frac{U}{A.\varepsilon_i.r_{i+1}}$$

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

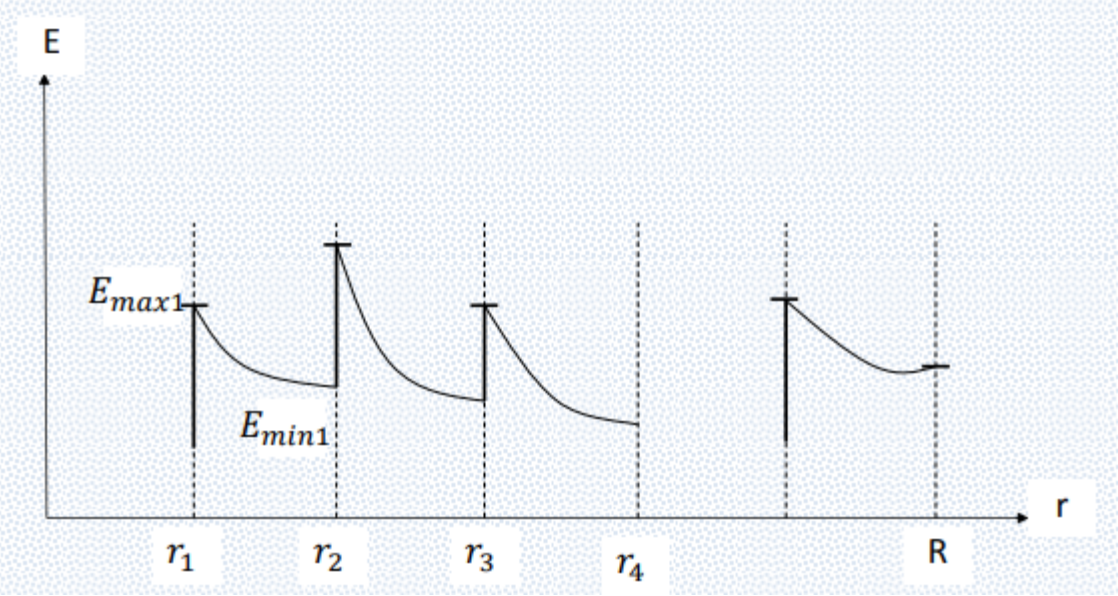
If the maximum electric field strength across any dielectric layer, $E_{i,max}$ becomes greater than breakdown strength of that dielectric material, there will be an electrical discharge.

$E_{i,max} \geq E_{bi} \rightarrow$ Discharge in that layer of the insulation!!!

Breakdown, short circuit in i^{th} dielectric layer

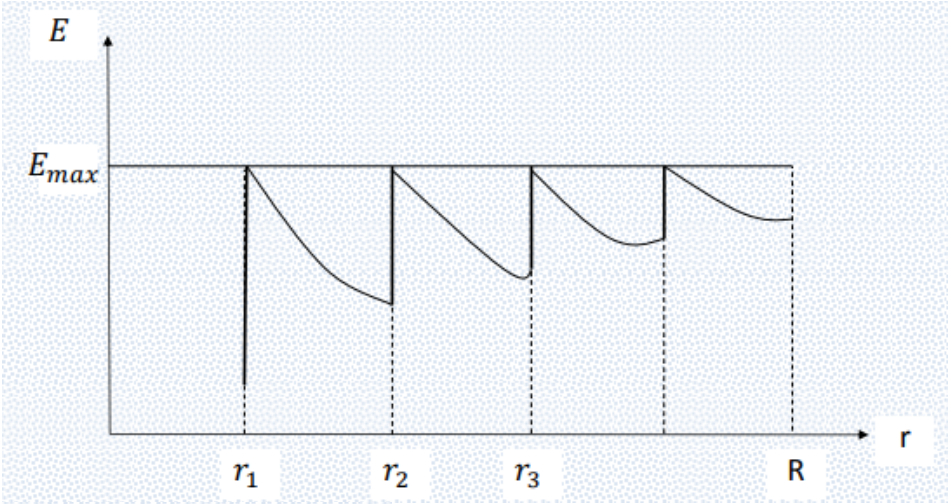
3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Electric field distribution:



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Uniform stress condition:



To make the field disribution more uniform; (uniformly stressed)

$$E_{1max} = E_{2max} = \cdots = E_{nmax}$$

$$\frac{U}{A.\varepsilon_1.l_1.r_1} = \frac{U}{A.\varepsilon_2.l_2.r_2} = \cdots = \frac{U}{A.\varepsilon_n.l_n.r_n} \rightarrow \varepsilon_1.l_1.r_1 = \varepsilon_2.l_2.r_2 = \cdots = \varepsilon_n.l_n.r_n$$

If $l_1 = l_2 = \cdots = l_n = l = constant$

$$\varepsilon_1.r_1 = \varepsilon_2.r_2 = \cdots = \varepsilon_n.r_n$$

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Actual electrode seperation, $d = r_2 - r_1$

$$E_{ave} = \frac{U}{d} \rightarrow d = \frac{U}{E_{ave}}$$

Effective electrode seperation, $\alpha = \frac{U}{E_{max}} = r_1 \ln \frac{r_2}{r_1}$

$$E_{max} = \frac{U}{\alpha} \rightarrow U = E_{max}.\alpha$$

Geometric characteristics, $p = \frac{r_1 + d}{r_1} = \frac{r_2}{r_1} = q$

Utilization factor (electrode efficiency), $\eta = \frac{E_{ave}}{E_{max}} = \frac{U/(r_2 - r_1)}{U/r_1 \ln \frac{r_2}{r_1}} = \frac{\alpha}{d} = \frac{r_1.lnp}{r_2 - r_1} = \frac{1}{p-1} lnp$

4 BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Analyzing the relationship between electrode radius, electric field and potential:

- Change in E_{max} with respect to r_1
- Change in U_{max} with respect to r_1

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in E_{max} with respect to r_1

Keeping r_2 and U constant

Radius of outer cylinder is constant, radius of inner cylinder is variable

$E_{max} = \frac{U}{r_1 ln\frac{r_2}{r_1}}$ E_{max} depends on r_1

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in E_{max} with respect to r_1

How can we find the minimum value for E_{max}? It is the case when its derivative is zero.

$$\frac{dE_{max}}{dr_1} = 0 \rightarrow$$

$$E_{max} = \frac{U}{r_1 . \ln\frac{r_2}{r_1}} = \frac{U}{r_1 . \ln r_2 - r_1 . \ln r_1}$$

$$\frac{dE_{max}}{dr_1} = \frac{-(\ln r_2 - \ln r_1 - 1).U}{(r_1 . \ln r_2 - r_1 . \ln r_1)^2} = 0$$

$$\ln r_2 - \ln r_1 - 1 = 0$$

$$\ln\frac{r_2}{r_1} = 1 = \ln e \rightarrow \frac{r_2}{r_1} = e$$

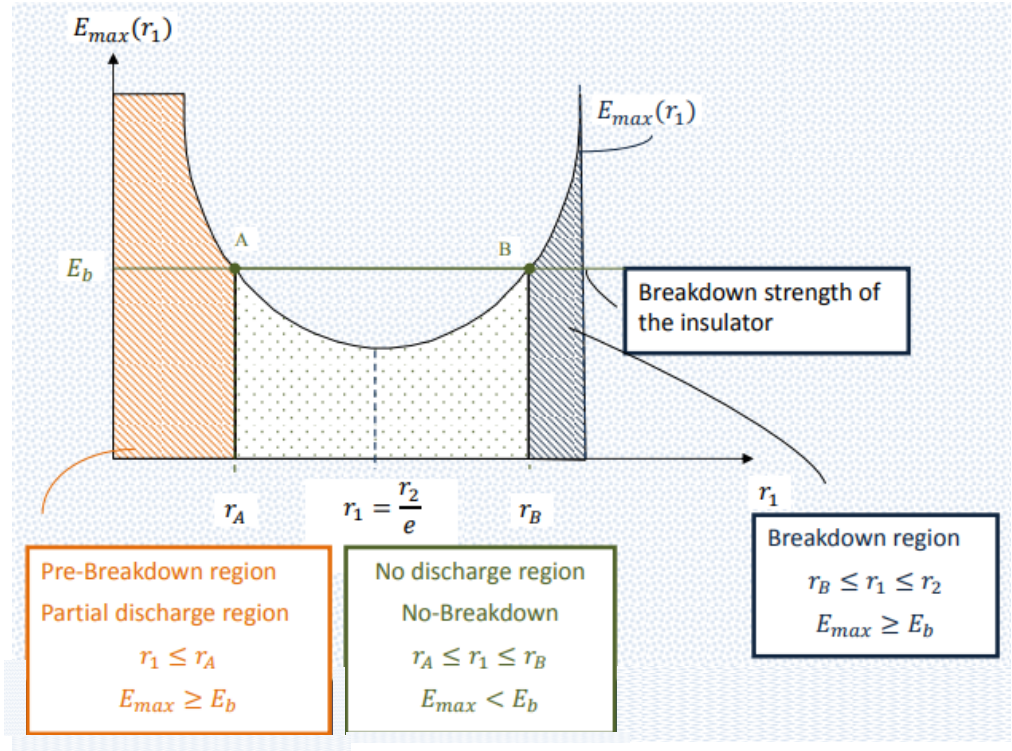
$$\frac{r_2}{r_1} = p = e = 2.718$$

$p = p_b$

fittest, optimum geometric char. with regard to breakdown in co-axial cylinders

4 BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in E_{max} with respect to r_1



BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in U_{max} with respect to r_1

Keeping r_2 and E_b constant

$$E_{max} = \frac{U}{r_1 ln \frac{r_2}{r_1}} = \frac{U}{\alpha}$$

$$U = \alpha . E_{max}$$

$\alpha = \alpha_{max}$ $\quad$ $U = U_{max}$ $\quad$ $E_{max} = E_b$

$$U_{max} = \alpha_{max} . E_b$$

$$\alpha = r_1 . \ln \frac{r_2}{r_1} = f(r_1)$$

Maximum ⟶ Voltage will be **Maximum**